INTERNATIONAL SEARCH REPORT

International application No
PCT/IB2022/052489

A. CLASSIFICATION OF SUBJECT MATTER
INV. G01N33/574 A61K39/00
ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)
G01N A61K

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)
EPO-Internal, BIOSIS, EMBASE

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	WO 2019/108900 A1 (NOVARTIS AG [CH]; UNIV PENNSYLVANIA [US]) 6 June 2019 (2019-06-06) page 2; page 29 et seq. -----	1-292
X	MURTA E F ET AL: "Lymphocyte Subpopulations in Patients with Advanced Breast Cancer Submitted to Neoadjuvant Chemotherapy", TUMORI., vol. 86, no. 5, 1 September 2000 (2000-09-01), pages 403-407, XP055928378, IT ISSN: 0300-8916, DOI: 10.1177/030089160008600507 the whole document ----- -/--	1-292

[X] Further documents are listed in the continuation of Box C.

[X] See patent family annex.

* Special categories of cited documents :

"A" document defining the general state of the art which is not considered to be of particular relevance

"E" earlier application or patent but published on or after the international filing date

"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance;; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance;; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
8 June 2022	21/06/2022

Name and mailing address of the ISA/	Authorized officer
European Patent Office, P.B. 5818 Patentlaan 2 NL - 2280 HV Rijswijk Tel. (+31-70) 340-2040, Fax: (+31-70) 340-3016	Rosin, Oliver

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Form PCT/ISA/210 (second sheet) (April 2005)

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PCT/IB2022/052489

C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	WO 2009/120341 A2 (UNIV SOUTH FLORIDA [US]; US DEPT VETERANS AFFAIRS [US] ET AL.) 1 October 2009 (2009-10-01) the whole document	1-292
X	WALAYAT SHAH ET AL: "A reversed CD4/CD8 ratio of tumor-infiltrating lymphocytes and a high percentage of CD4+FOXP3+ regulatory T cells are significantly associated with clinical outcome in squamous cell carcinoma of the cervix", CELLULAR & MOLECULAR IMMUNOLOGY, vol. 8, no. 1, 1 January 2010 (2010-01-01) , pages 59-66, XP055238962, London ISSN: 1672-7681, DOI: 10.1038/cmi.2010.56 the whole document	1-292
X	ALFRED L. GARFALL ET AL: "T-cell phenotypes associated with effective CAR T-cell therapy in postinduction vs relapsed multiple myeloma", BLOOD ADVANCES, vol. 3, no. 19, 8 October 2019 (2019-10-08), pages 2812-2815, XP055768817, ISSN: 2473-9529, DOI: 10.1182/bloodadvances.2019000600 the whole document	1-292
X	ZUAZO MIREN ET AL: "Systemic CD4 Immunity as a Key Contributor to PD-L1/PD-1 Blockade Immunotherapy Efficacy", FRONTIERS IN IMMUNOLOGY, vol. 11, 30 November 2020 (2020-11-30), XP055928388, DOI: 10.3389/fimmu.2020.586907 Retrieved from the Internet: URL:https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7734243/pdf/fimmu-11-586907.pdf> the whole document	1-292
T	WO 2021/168376 A1 (KITE PHARMA INC [US]) 26 August 2021 (2021-08-26)	

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Box No. I Nucleotide and/or amino acid sequence(s) (Continuation of item 1.c of the first sheet)

1. With regard to any nucleotide and/or amino acid sequence disclosed in the international application, the international search was carried out on the basis of a sequence listing:

   a. [X] forming part of the international application as filed:

      [X] in the form of an Annex C/ST.25 text file.

      [ ] on paper or in the form of an image file.

   b. [ ] furnished together with the international application under PCT Rule 13ter.1(a) for the purposes of international search only in the form of an Annex C/ST.25 text file.

   c. [ ] furnished subsequent to the international filing date for the purposes of international search only:

      [ ] in the form of an Annex C/ST.25 text file (Rule 13ter.1(a)).

      [ ] on paper or in the form of an image file (Rule 13ter.1(b) and Administrative Instructions, Section 713).

2. [ ] In addition, in the case that more than one version or copy of a sequence listing has been filed or furnished, the required statements that the information in the subsequent or additional copies is identical to that forming part of the application as filed or does not go beyond the application as filed, as appropriate, were furnished.

3. Additional comments:

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No
PCT/IB2022/052489

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
WO 2019108900 A1	06-06-2019	AU 2018375738 A1	11-06-2020
		BR 112020010579 A2	10-11-2020
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		EP 3717907 A1	07-10-2020
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		US 2022169694 A1	02-06-2022
		WO 2021168376 A1	26-08-2021